

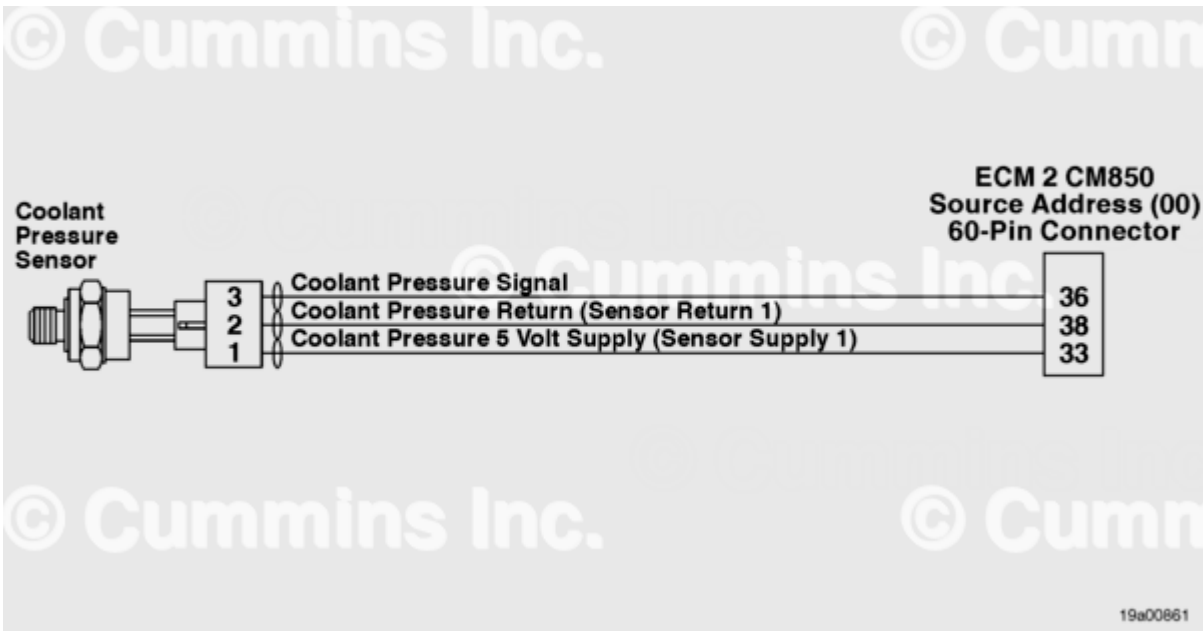
# Fault Code: 228

## Coolant Pressure - Data Valid But Below Normal Operating Range - Most Severe Level

 Printable Version

### Overview

| Codes                                                                                                   | Reason                                                                                                   | Effect                      |
|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------|
| <p>Fault Code: 228</p> <p>PID(P): P109</p> <p>SPN: 109</p> <p>FMI: 1/1</p> <p>Lamp: Red</p> <p>SRT:</p> | <p>Coolant pressure signal indicates coolant pressure is below the engine protection critical limit.</p> | <p>Engine may shutdown.</p> |



Coolant Pressure Sensor Circuit

### Circuit Description

The engine coolant pressure sensor is a variable resistance sensor used by the ECM to monitor the coolant pressure. The engine coolant pressure sensor has three circuits: 5 volt supply, return and signal circuits. The signal circuit voltage indicates the coolant pressure in the engine cooling system.

## Component Location

The coolant pressure sensor is located on the water pump connection to the engine block.

## Conditions For Running The Diagnostics

This diagnostic runs continuously when the engine is running.

## Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected the engine coolant pressure was less than the engine protection limit.

## Action Taken When The Fault Code Is Active

- The generator set controller displays a shutdown fault immediately when the diagnostics runs and fails.
- The engine will be shut down if the Engine Protection Shutdown feature is enabled.
- The torque output of the engine will be reduced.

## Conditions For Clearing The Fault Code

- To validate the repair, operate the engine under load for at least 5 minutes.
- The generator set controller will turn off the shutdown indicator immediately after the user presses the reset.
- The "Reset All Faults" command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

## Shoptalk

Possible causes of this fault code include:

- Low coolant level
- A malfunctioning radiator cap.
- A malfunctioning coolant pressure sensor.
- A malfunctioning water pump.

Refer to Troubleshooting Fault Code 228 . (</qs3/pubsys2/xml/en/procedures/60/60-t05-228.html>)

